

Homework 1 - The Learning Procedure

Parsa Delavari

Due 23 September at 11pm

Instructions

- Use the space inside of

```
::: {.solbox data-latex=""}
```

```
:::
```

to answer the following questions.

- Do not move this file or copy it and work elsewhere. Work in the same directory.
- Use a new branch named whatever you want. Create it now! Can't come up with something, try here. Make a small change, say by adding your name. Commit and push now!
- Try to Knit your file now. Are there errors? Fix them now, not at 11pm on the due date.
- There MUST be some text between `::: {.solbox}` and the next `:::` or this will fail to Knit.
- If your code or figures run off the edge of the `.pdf`, you'll lose 2 points automatically.
- Be sure to add your name in the `author` field at the top.

Generative AI Self Report

If you use any generative AI tool on homeworks or labs, indicate here how you used the tools and how they contributed to your learning/ability to understand the assignment. Include every prompt that you used.

This self report will not affect your grade in any way. Please be as honest as possible.

- your response here *

1 Forecasting COVID in BC

In the course `Stat406` package, you will find COVID-19 case data for the five health authorities in BC. We're going to play with a few models for making forecasts. Your goal is to **predict 2-week ahead case counts for all 5 health authorities**.

There are 2 BC Covid datasets we need for this assignment: `bccovid_train` and `bccovid_test`. Both have 6 variables: `HA`, `date`, `cases+14`, `cases+0`, `cases-7`, `cases-14`. `cases+14` is the response: it is the total number of cases observed in that specific HA, in the week ending 14 days after `date`. The other `cases` variables are features: the total number of cases in the week ending on the date, those for 1 week earlier and those for 2 weeks earlier. For complete descriptions, try `?bccovid` after loading the course package (or examine the package documentation on the web).

1.1 (1 point)

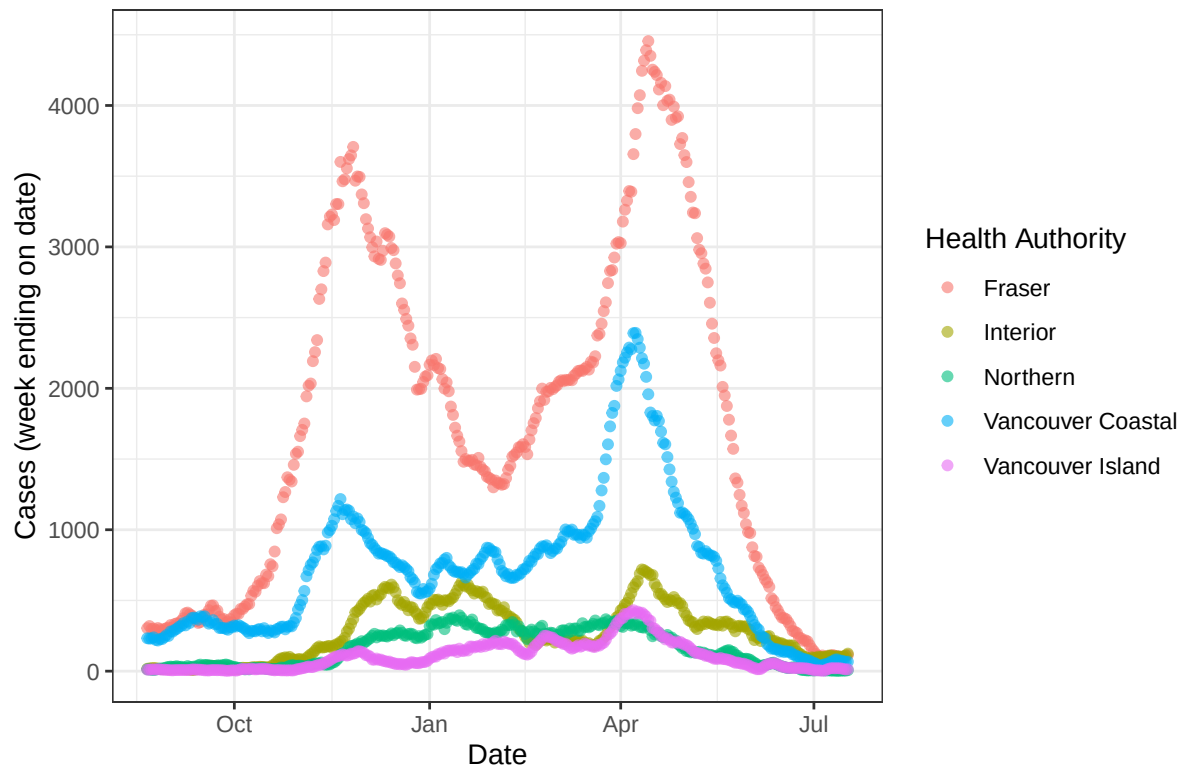
Using only the training data: a. (0.5 point) make a plot of today's cases over time. colour the points by health authority. b. (0.5 point) Write a few sentences discussing the data. Describe any similarities or differences across health authorities you notice. How do cases seem to "evolve" over time?

Hint: to use these covariates in `ggplot()`, you have to put them inside "backtick" marks, like ``cases+0``.

a.

```
ggplot(bccovid_train) +  
  geom_point(alpha = 0.6, aes(x = date, y = `cases+0`, colour = HA)) +  
  labs(title = "Weekly COVID-19 Cases in BC Health Authorities",  
        x = "Date",  
        y = "Cases (week ending on date)",  
        colour = "Health Authority")
```

Weekly COVID-19 Cases in BC Health Authorities



b.

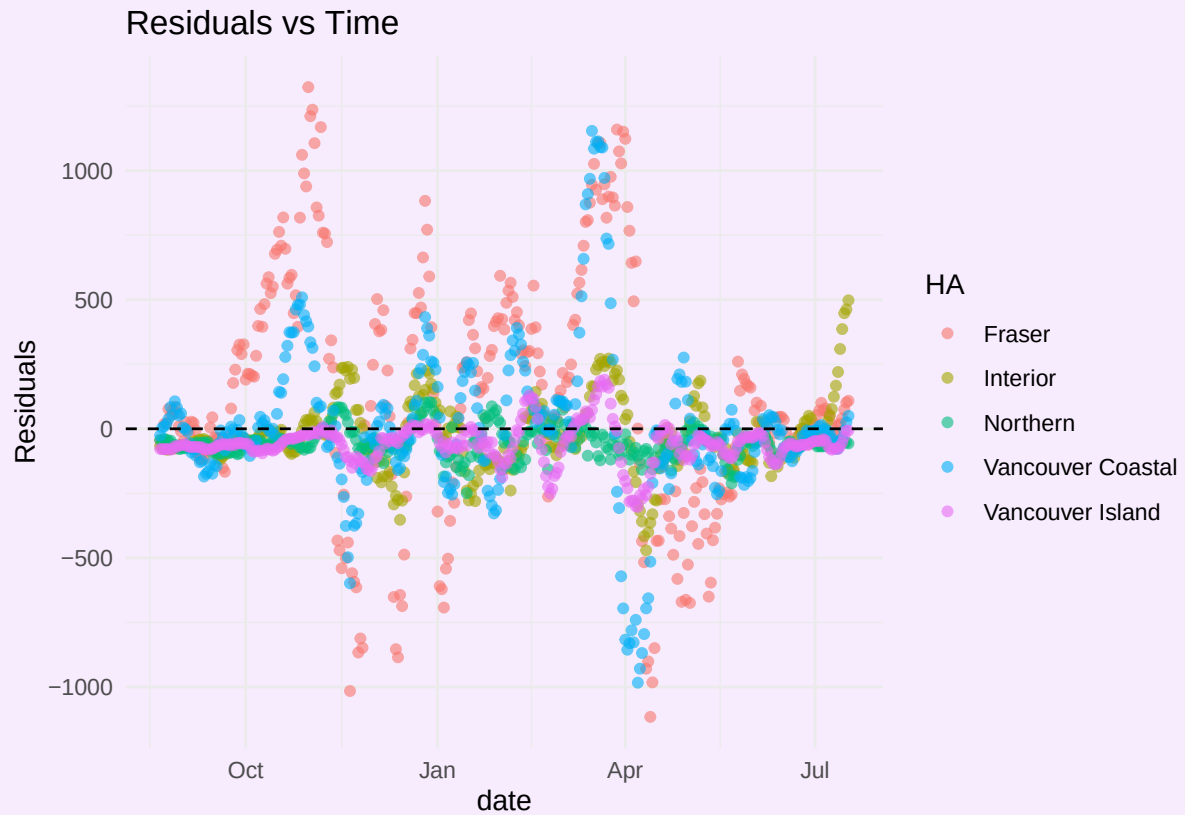
1.2 (2 points)

Estimate a linear model of 2-week ahead cases (`cases+14`) on the three other case features using the training data.

- (0.25 points) Plot the residuals against time (again coloured by HA).
- (0.5 points) Make a QQ-plot of the residuals (coloured by HA). Do these plots seem support the standard assumptions for a linear model? Why or why not?
- (0.5 point) Report the slope coefficient on `cases+0`. Describe what this coefficient means in the context of this problem. Does this interpretation make sense, why or why not?
- (0.75 points) Does this model seem appropriate? Justify your answer.

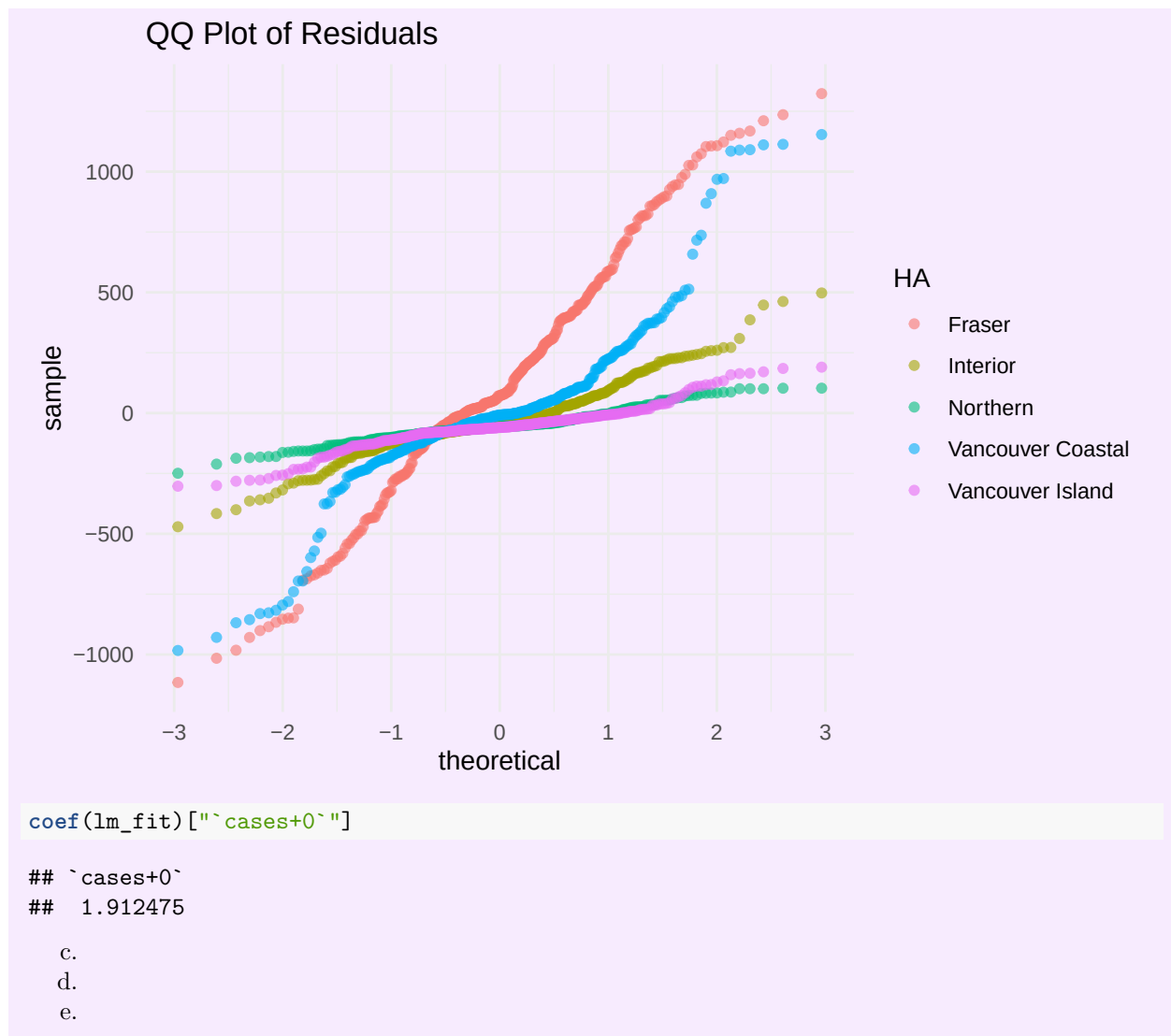
Hint: to use these covariates in a formula, you have to put them inside “backtick” marks, like ``cases+0``.

```
a.  
# Fit linear model  
lm_fit <- lm(`cases+14` ~ `cases+0` + `cases-7` + `cases-14`, data = bccovid_train)  
  
length(residuals(lm_fit))  
  
## [1] 1655  
  
bccovid_train$resid <- residuals(lm_fit)  
  
ggplot(bccovid_train, aes(x = date, y = resid, colour = HA)) +  
  geom_point(alpha = 0.6) +  
  geom_hline(yintercept = 0, linetype = "dashed") +  
  labs(title = "Residuals vs Time", y = "Residuals") +  
  theme_minimal()
```



b.

```
ggplot(bccovid_train, aes(sample = resid, colour = HA)) +
  stat_qq(alpha = 0.6) +
  labs(title = "QQ Plot of Residuals") +
  theme_minimal()
```



1.3 (2 points)

Examine the `bccpop` data set. Using an appropriate `*_join()`, add this column to the training data. Now scale all the columns by the population to get “cases per capita”. Multiply by 100,000 just to get reasonable looking numbers.

- (0.5 point) Make a plot of today’s cases over time scaled by population (and coloured by HA). Furthermore, scale the y-axis on the `log10()` scale. (Hint: try `?scale_y_log10`). How does this compare to the plot in Question 1.1? Write 2 complete sentences.
- (1.5 points) Repeat Question 1.2 with both your scaled data after taking `log()` of all the predictors AND using HA as a categorical predictor.

First bullet.

```
bccovid_train2 <- bccovid_train %>%
  left_join(bccpop, by = "HA")

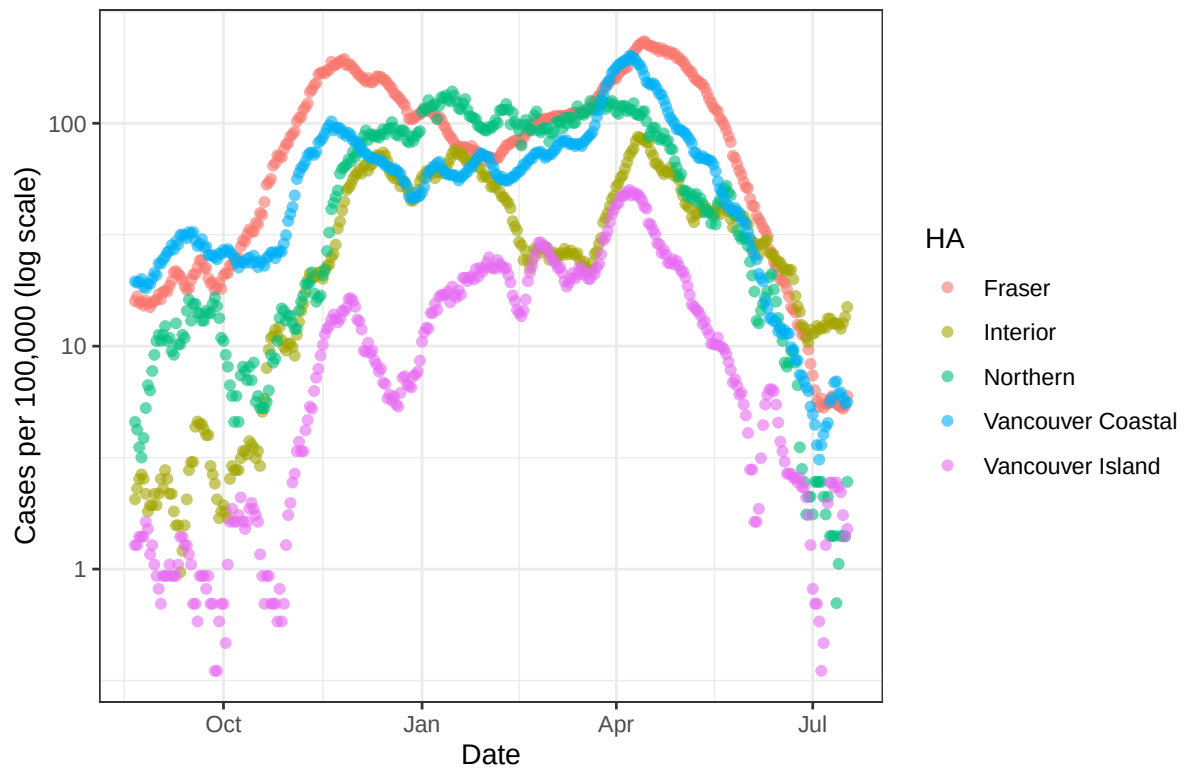
bccovid_train2

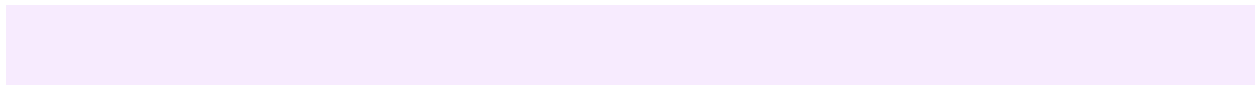
## # A tibble: 1,655 x 8
##   HA      date      `cases+14` `cases+0` `cases-7` `cases-14` resid    pop
##   <chr> <date>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl> <int>
## 1 Fraser 2020-08-21      339      304      269      185 -46.1 1906933
## 2 Fraser 2020-08-22      335      319      277      172 -72.8 1906933
## 3 Fraser 2020-08-23      356      298      311      162  18.6 1906933
## 4 Fraser 2020-08-24      374      301      298      180  20.8 1906933
## 5 Fraser 2020-08-25      409      290      296      196  76.7 1906933
## 6 Fraser 2020-08-26      414      305      292      212  51.0 1906933
## 7 Fraser 2020-08-27      402      296      308      222  72.0 1906933
## 8 Fraser 2020-08-28      393      285      304      269  85.4 1906933
## 9 Fraser 2020-08-29      378      298      319      277  60.1 1906933
## 10 Fraser 2020-08-30      355      304      298      311   9.93 1906933
## # i 1,645 more rows

bccovid_train2 <- bccovid_train2 %>%
  mutate(`cases+0_pc` = (`cases+0` / pop) * 100000,
         `cases-7_pc` = (`cases-7` / pop) * 100000,
         `cases-14_pc` = (`cases-14` / pop) * 100000,
         `cases+14_pc` = (`cases+14` / pop) * 100000)

ggplot(bccovid_train2, aes(x = date, y = `cases+0_pc`, colour = HA)) +
  geom_point(alpha = 0.6) +
  scale_y_log10() +
  labs(title = "Per-capita weekly cases (per 100,000) in BC",
       y = "Cases per 100,000 (log scale)",
       x = "Date")
```

Per-capita weekly cases (per 100,000) in BC

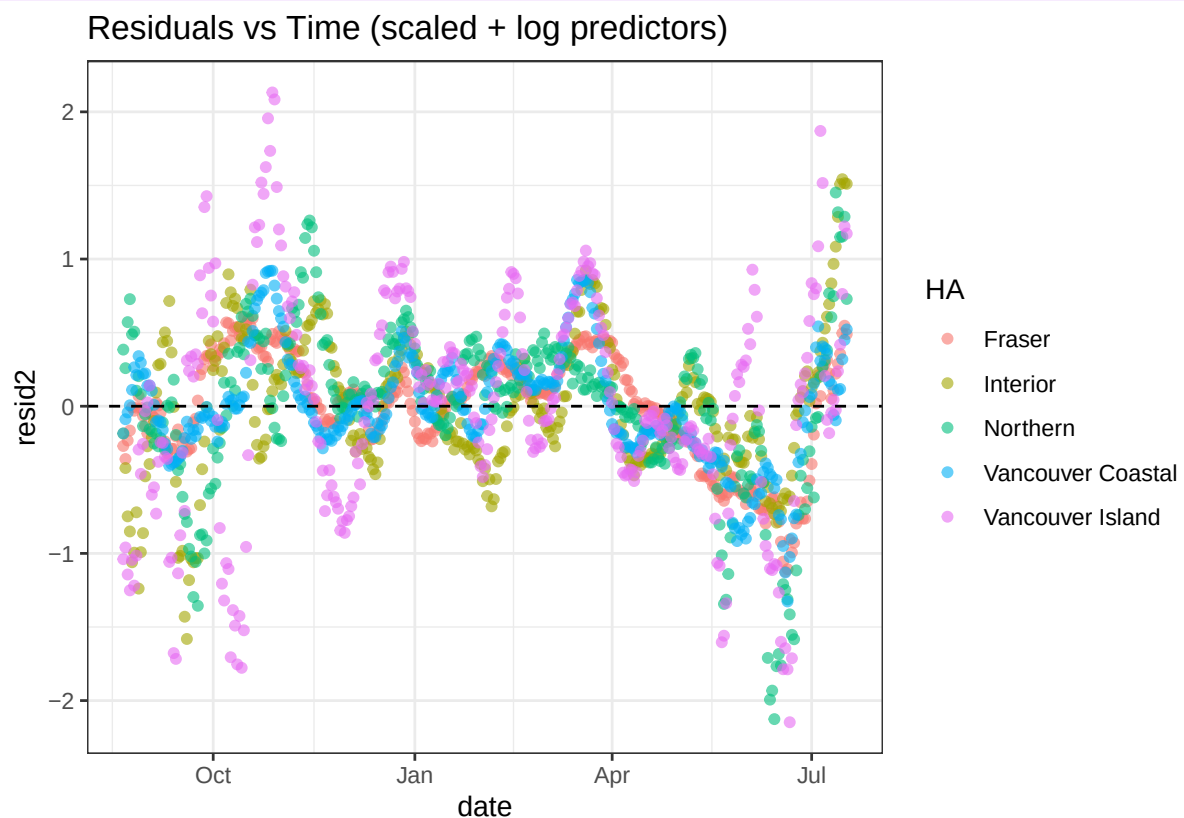




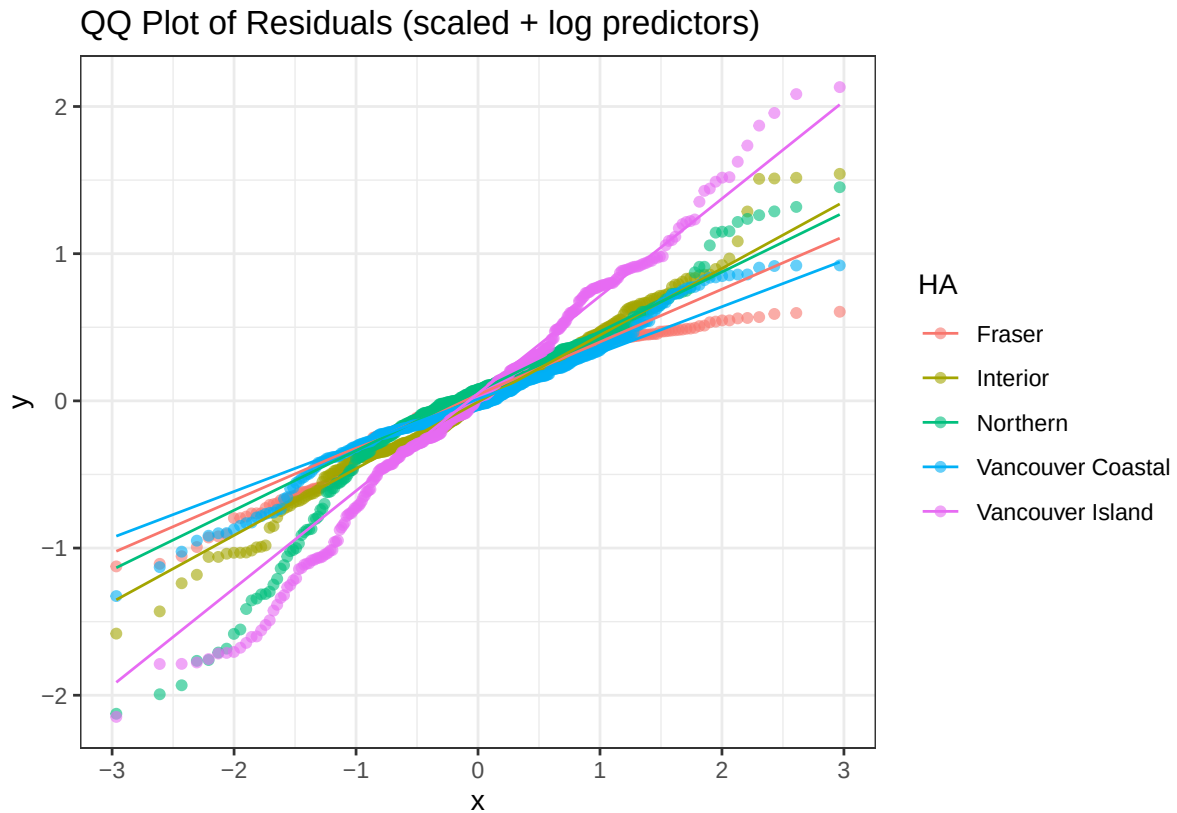
Repeat Q2 with different data.

a.

```
lm_fit2 <- lm(log(`cases+14_pc`) ~ log(`cases+0_pc`) +  
              log(`cases-7_pc`) +  
              log(`cases-14_pc`) +  
              HA,  
              data = bccovid_train2)  
  
bccovid_train2$resid2 <- residuals(lm_fit2)  
  
# Residuals vs time  
ggplot(bccovid_train2, aes(x = date, y = resid2, colour = HA)) +  
  geom_point(alpha = 0.6) +  
  geom_hline(yintercept = 0, linetype = "dashed") +  
  labs(title = "Residuals vs Time (scaled + log predictors)")
```



```
# QQ plot  
ggplot(bccovid_train2, aes(sample = resid2, colour = HA)) +  
  stat_qq(alpha = 0.6) +  
  stat_qq_line() +  
  labs(title = "QQ Plot of Residuals (scaled + log predictors)")
```



- b.
- c.
- d.
- e.

1.4 (1 point)

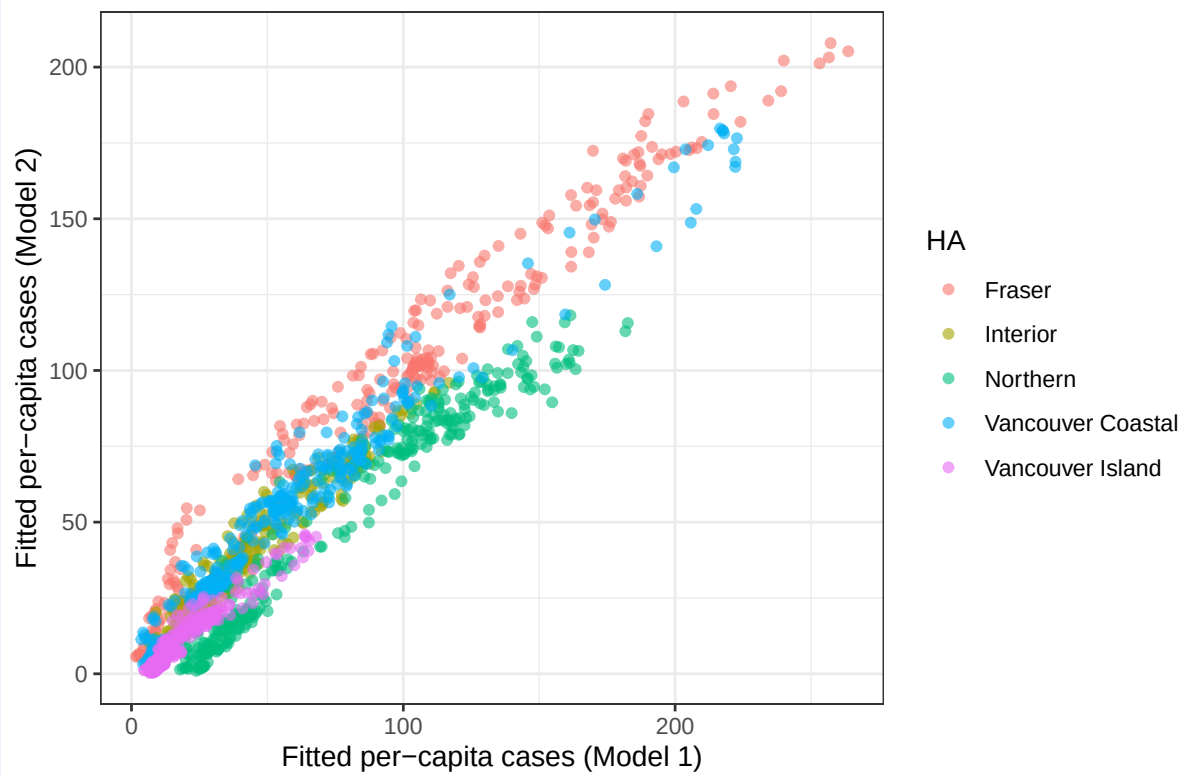
Plot the fitted values from the two models against each other (fits from the Question 1.2 on the x-axis, with fits from Question 1.3 on the y-axis). Be sure to transform them as needed to be comparable. Colour the points by HA. What do you notice?

```
# fitted values
bccovid_train2$fit1 <- fitted(lm_fit) # from Q1.2 (raw counts)

bccovid_train2$fit2 <- exp(fitted(lm_fit2)) # from Q1.3 (log per capita)
bccovid_train2$fit1_pc <- (bccovid_train2$fit1 / bccovid_train2$pop) * 100000

ggplot(bccovid_train2, aes(x = fit1_pc, y = fit2, colour = HA)) +
  geom_point(alpha = 0.6) +
  labs(title = "Fitted values: raw vs log-scaled models",
       x = "Fitted per-capita cases (Model 1)",
       y = "Fitted per-capita cases (Model 2)")
```

Fitted values: raw vs log-scaled models



(placeholder text)

1.5 (1 point)

Calculate the root mean squared error of both models. Discuss the results in 2 sentences.

```
rmse1 <- sqrt(mean(residuals(lm_fit)^2))

# observed per-capita
obs_pc <- (bccovid_train2$`cases+14` / bccovid_train2$pop) * 100000

# fitted per-capita (already back-transformed earlier)
fit2 <- exp(fitted(lm_fit2))

rmse2 <- sqrt(mean((obs_pc - fit2)^2))

print(rmse1)

## [1] 269.2422

print(rmse2)

## [1] 20.40529

(placeholder text)
```

1.6 (1 point)

Use the K-fold CV function from class (loaded below) to compare these two models (the first model being the one with `bccovid_train` from Question 1.2, and the second being the one with the log of case rates and the factor for HA from Question 1.3). Use $K = 10$. In order to do this, you'll need to be sure that the errors are on the same scale.

Use `Cases` as the scale. This means you'll have to specify the model and the error function carefully.

- (0.5 points) What are the CV scores for both models?
- (0.25 points) Which do you prefer?
- (0.25 points) How much more accurate is it?

```
#' @param data The full data set
#' @param estimator Function. Has 1 argument (some data) and fits a model.
#' @param predictor Function. Has 2 args (the fitted model, the_newdata) and
#'   produces predictions
#' @param error_fun Function. Has one arg: the test data, with fits added.
#' @param kfolds Integer. The number of folds.
kfold_cv <- function(data, estimator, predictor, error_fun, kfolds = 5) {
  n <- nrow(data)
  fold.labels <- sample(rep(1:kfolds, length.out = n))
  errors <- double(kfolds)
  for (fold in 1:kfolds) {
    test.rows <- fold.labels == fold
    train <- data[!test.rows, ]
    test <- data[test.rows, ]
    current_model <- estimator(train)
    test$.preds <- predictor(current_model, test)
    errors[fold] <- error_fun(test)
  }
}
```

```
mean(errors)
}
```

```
error_fun <- function(test) {
  sqrt(mean((test$`cases+14` - test$.preds)^2, na.rm = TRUE))
}

estimator1 <- function(train) {
  lm(`cases+14` ~ `cases+0` + `cases-7` + `cases-14`, data = train)
}

predictor1 <- function(model, newdata) {
  predict(model, newdata)
}

estimator2 <- function(train) {
  lm(log(`cases+14` / pop) * 100000) ~
    log(`cases+0` / pop) * 100000 +
    log(`cases-7` / pop) * 100000 +
    log(`cases-14` / pop) * 100000 +
    HA,
  data = train)
}

predictor2 <- function(model, newdata) {
  # Predict in log-per-capita space
  pred_log <- predict(model, newdata)

  # Back-transform to per-capita
  pred_pc <- exp(pred_log)

  # Convert back to raw counts
  (pred_pc / 100000) * newdata$pop
}

set.seed(123) # for reproducibility

cv1 <- kfold_cv(bccovid_train2, estimator1, predictor1, error_fun, kfolds = 10)
cv2 <- kfold_cv(bccovid_train2, estimator2, predictor2, error_fun, kfolds = 10)

cv1

## [1] 270.5449

cv2

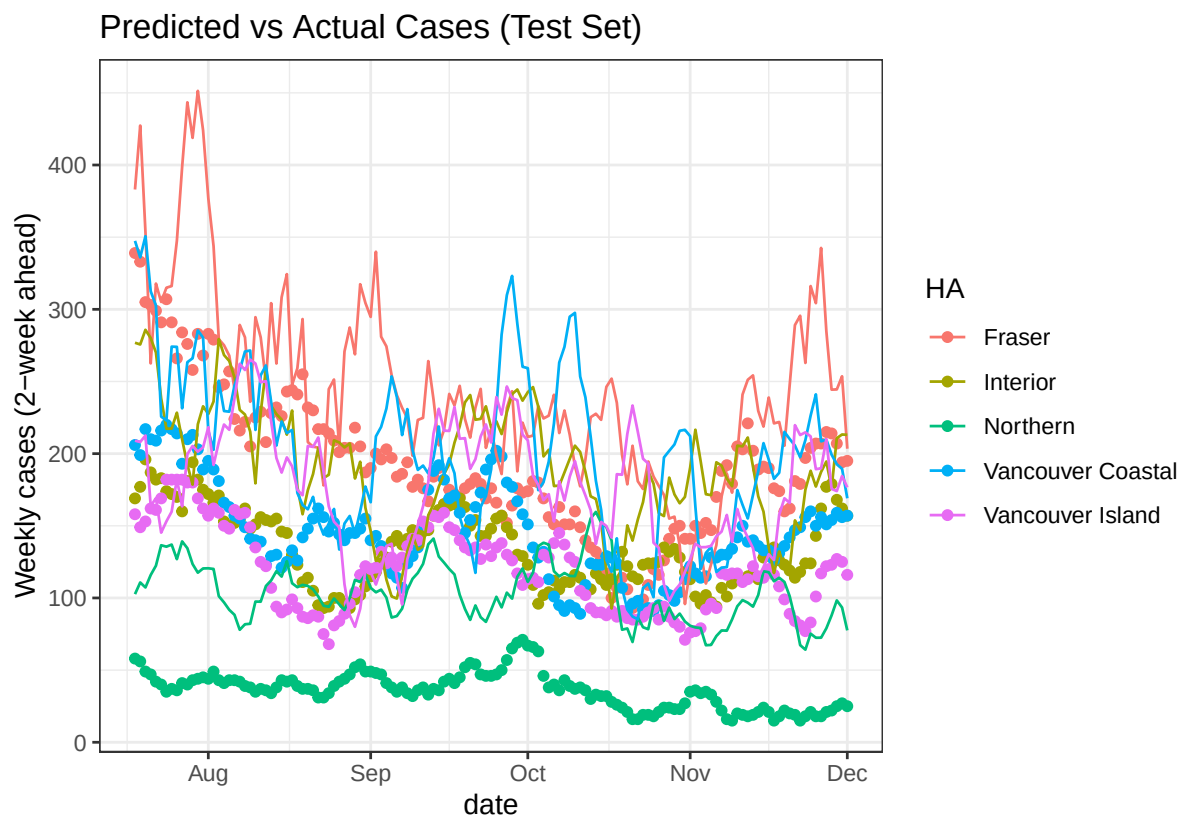
## [1] 278.4142

a.
b.
c.
```

1.7 (1 point)

Use your preferred model to predict the test set. a. (0.25 points) Plot the actual values and the predictions, coloured by HA. Use lines for the predictions and dots for the observations. b. (0.25 points) For each HA, how accurate is the prediction on average (what is the mean absolute error on the scale of cases)? c. (0.25 points) What about overall? d. (0.25 points) How does this compare to the CV score you computed above?

```
a.  
model_pref <- lm(`cases+14` ~ `cases+0` + `cases-7` + `cases-14`, data = bccovid_train)  
pred_raw <- predict(model_pref, bccovid_test)  
bccovid_test$.preds <- pred_raw  
  
library(ggplot2)  
ggplot(bccovid_test, aes(x = date, colour = HA)) +  
  geom_point(aes(y = `cases+14`)) +           # actuals  
  geom_line(aes(y = .preds)) +                # predictions  
  labs(title = "Predicted vs Actual Cases (Test Set)",  
        y = "Weekly cases (2-week ahead)")
```



```

# MAE per HA
mae_by_HA <- bccovid_test %>%
  group_by(HA) %>%
  summarise(mae = mean(abs(`cases+14` - .preds), na.rm = TRUE))

# overall MAE
mae_overall <- sqrt(mean(abs(bccovid_test$`cases+14` - bccovid_test$.preds)^2, na.rm = TRUE))

mae_by_HA

## # A tibble: 5 x 2
##   HA          mae
##   <chr>      <dbl>
## 1 Fraser      57.8
## 2 Interior    60.6
## 3 Northern    68.0
## 4 Vancouver Coastal 59.6
## 5 Vancouver Island  60.6

mae_overall

## [1] 71.53737

b.
c.
d.

```


1.8 (2 points)

- a. (0.25 points) Write the statistical model used for Question 1.3 (the log LM), and write the distributional assumption on the model residuals.
- b. (0.5 points) Assume we now modify the distribution on the residuals to be i.i.d. draws from Laplace distribution with μ (location parameter) = 0 and b (scale parameter) = 1. I claim that this model is more robust to outliers in the data. Why? Limit your answer to two sentences. (Hint: Plot the Laplace distribution and the Normal distribution side-by-side, and compare the tail behavior and probability density for large deviations from the mean.)
- c. (0.5 points) Derive the MLE for this model. This estimator does not have a closed form, so your final answer should be expressed as an optimization problem (i.e. argmax or argmin) over a summation, just like what we derived in class for logistic regression. Show your work for taking the derivative and setting it to zero, even though you cannot solve it explicitly.
- d. (0.25 points) What is the continuous risk function that, if we were to take an ERM perspective, yields the same estimator as the MLE?
- e. (0.5 points) Without actually fitting the model, give one (~2 sentence reason) why the Laplace model might perform better than the existing log LM for the COVID data. Give one (~2 sentence reason) why the existing model might perform better. Base your answers on the graphs and metrics you have computed so far.

- a.
- b.
- c.
- d.
- e.